



Intelligent Systems and
Analytics - ISA
fortnightly seminar



3D Point Cloud Reconstruction Algorithms for Plant Phenotyping

3 September ,2026

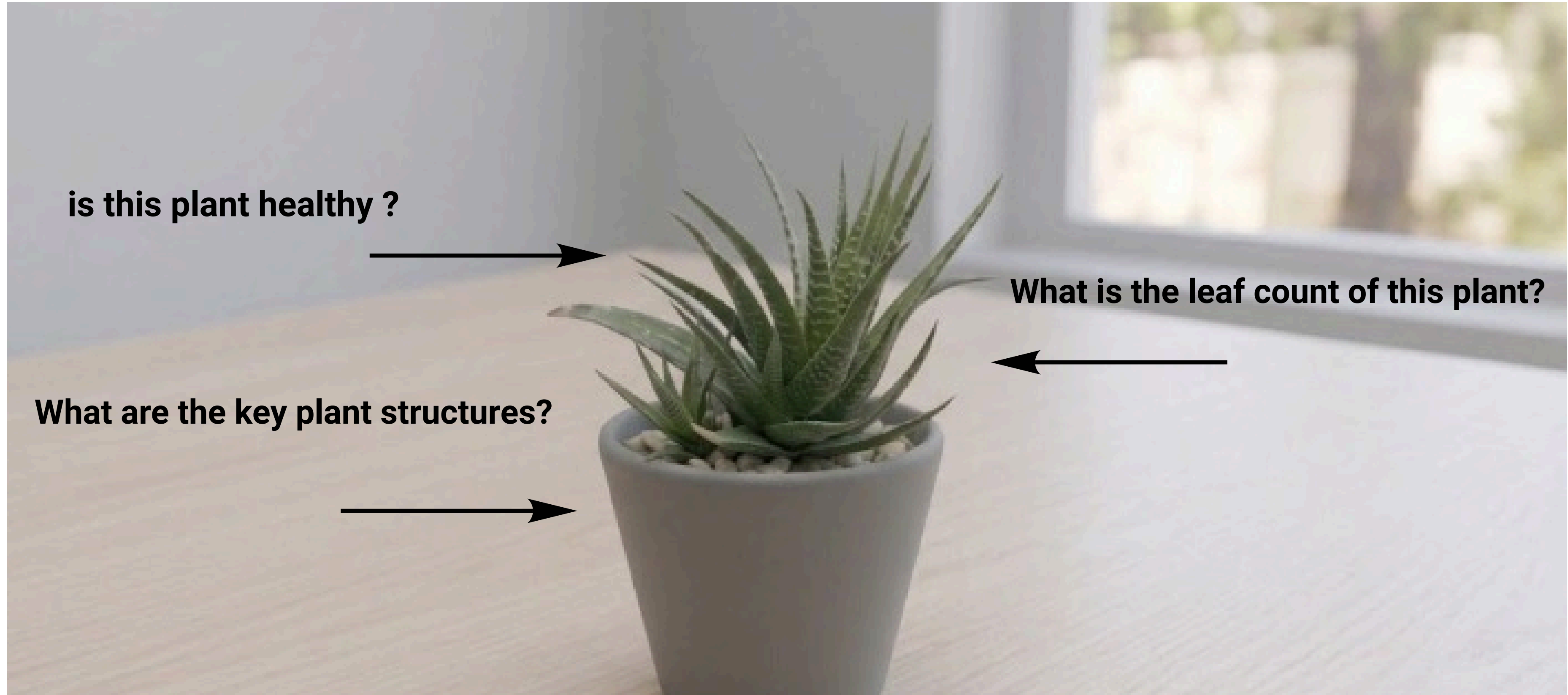
presented by : Aymen Turki

supervised by : Prof.Faouzi Alaya Cheikh

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Introduction



is this plant healthy ?



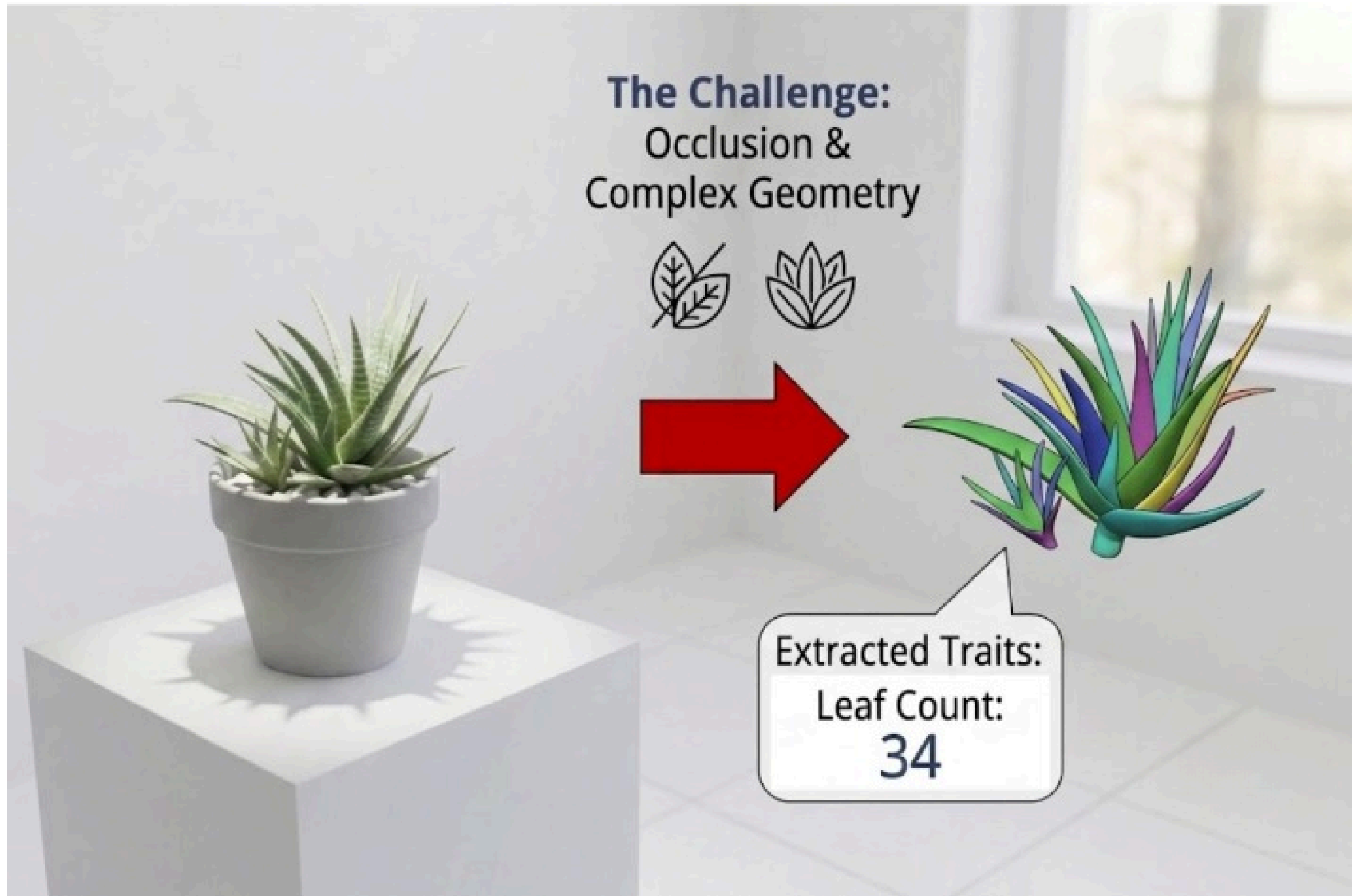
What is the leaf count of this plant?



What are the key plant structures?



Problematic



Proposed Solution:

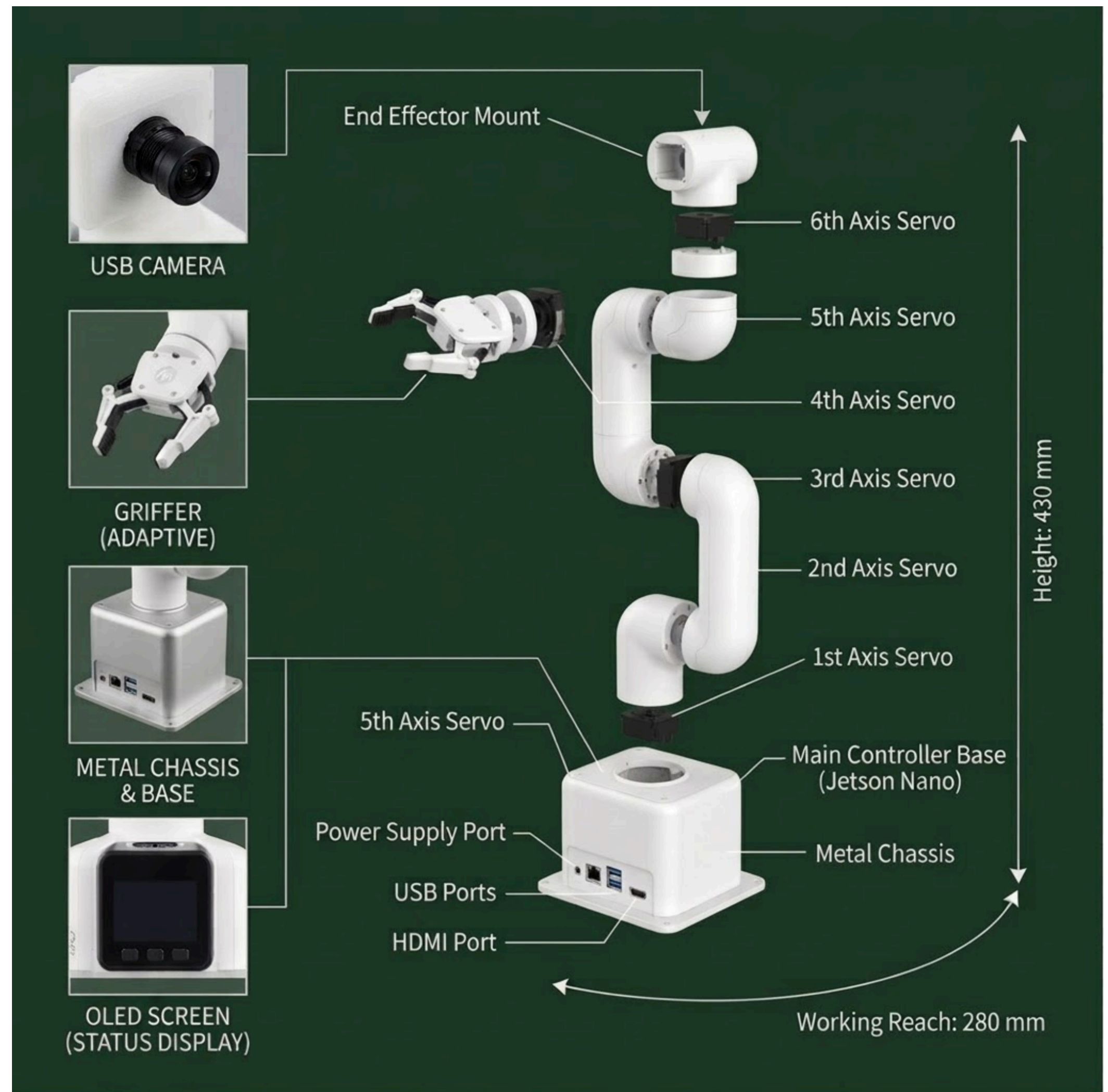
Automated Robotic 3D Plant Acquisition & Reconstruction



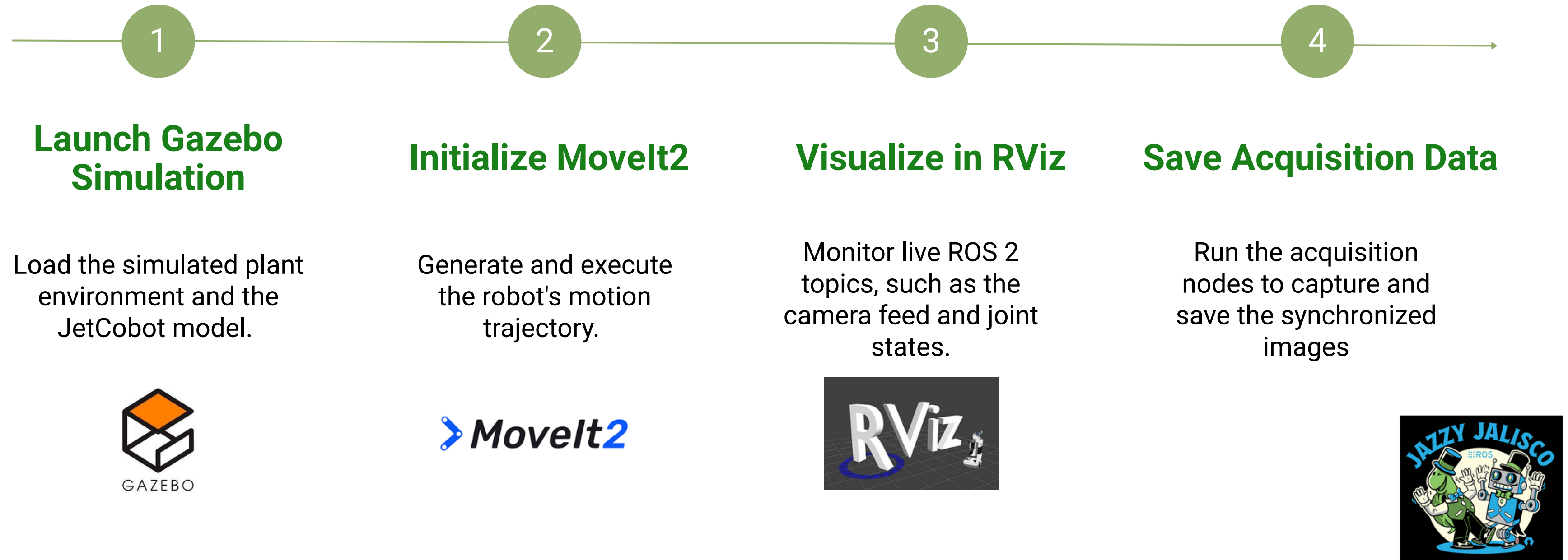
JetCobot 7-Axis Robotic Arm

Methodology Advantages

- Vs. Fixed 2D: Multi-view scanning eliminates self-occlusion.
- Vs. Heavy/Expensive LiDAR Rigs: Low-cost, lightweight RGB with rich visual textures.

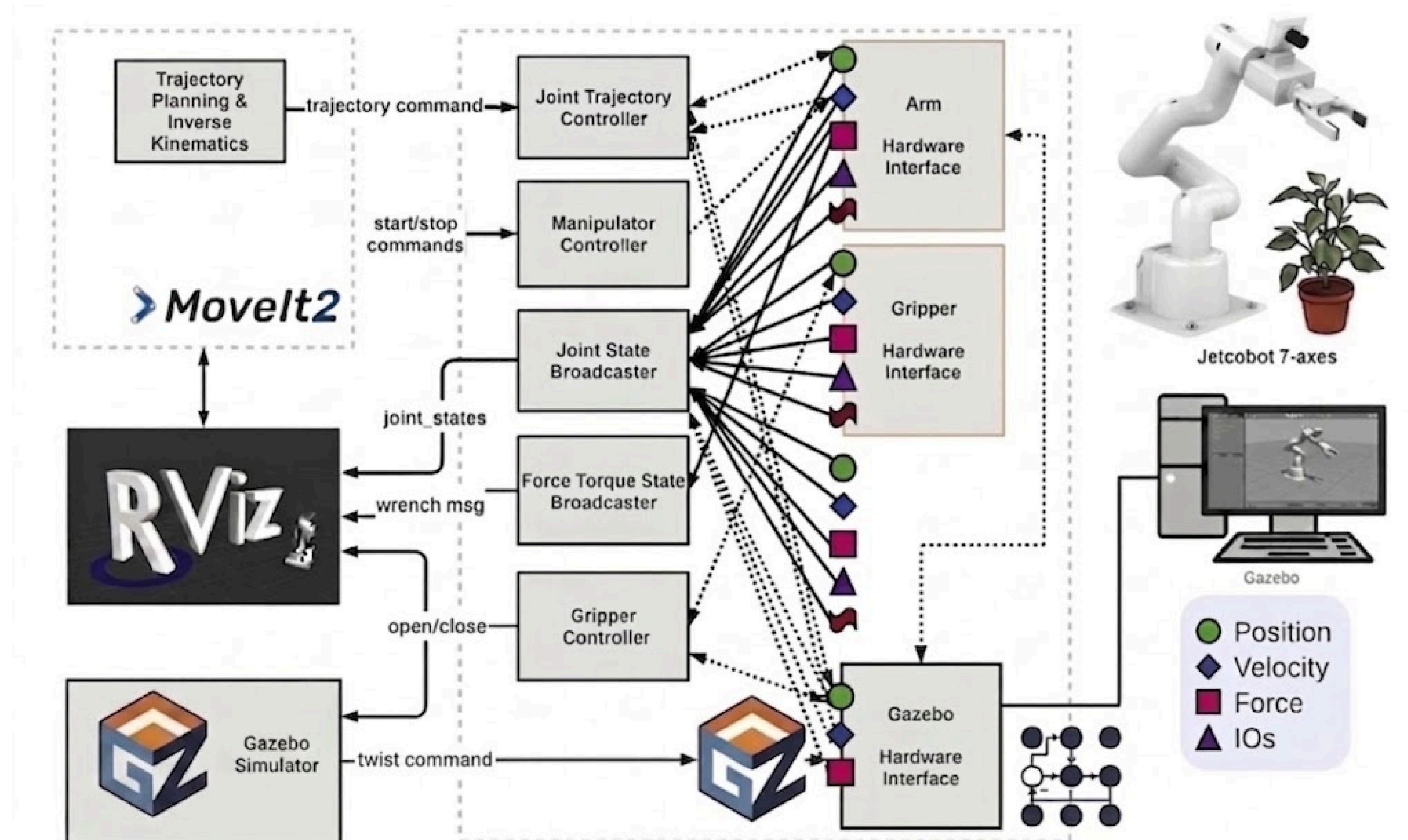


ROS 2 Control & Simulation Stack

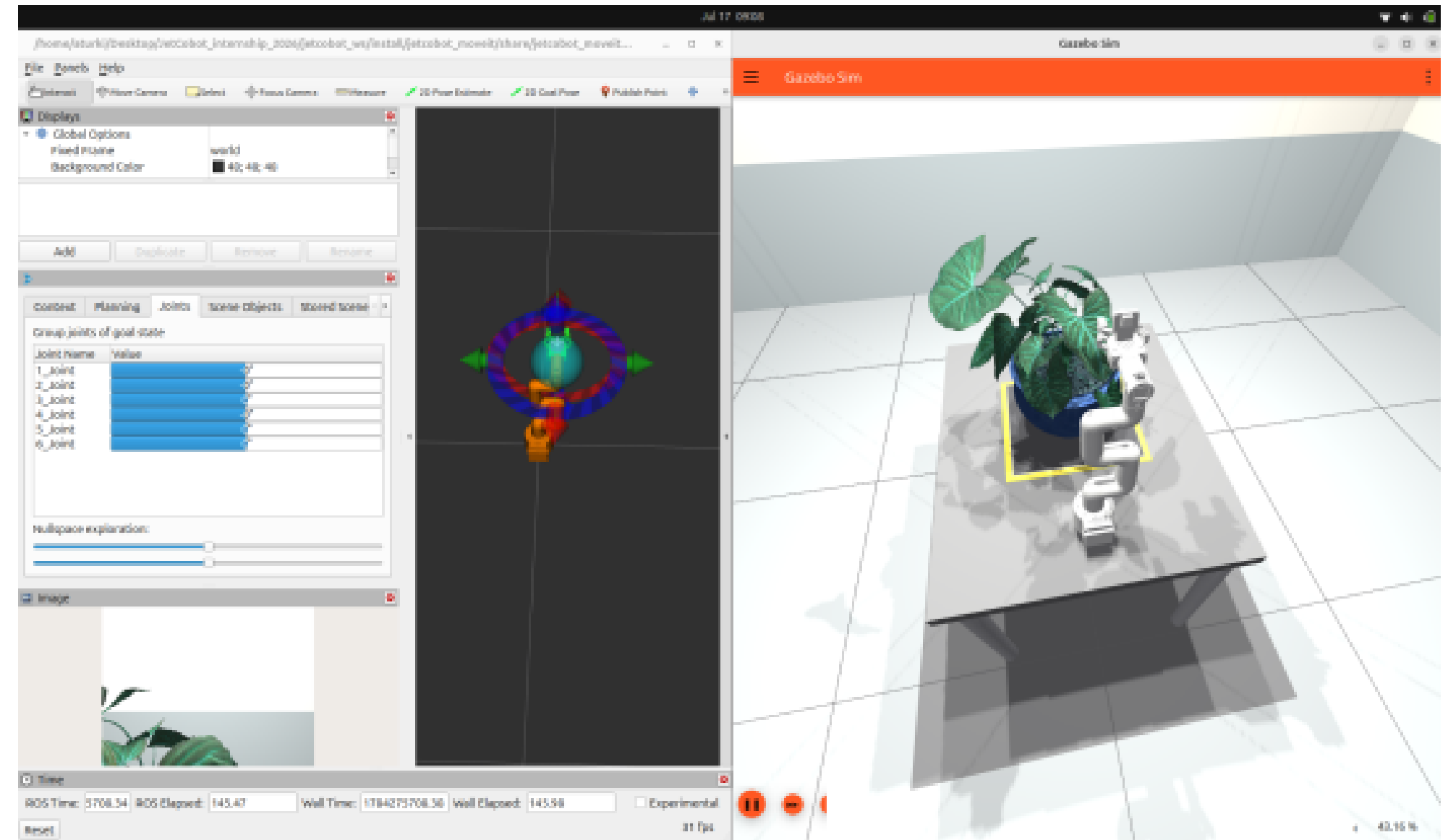
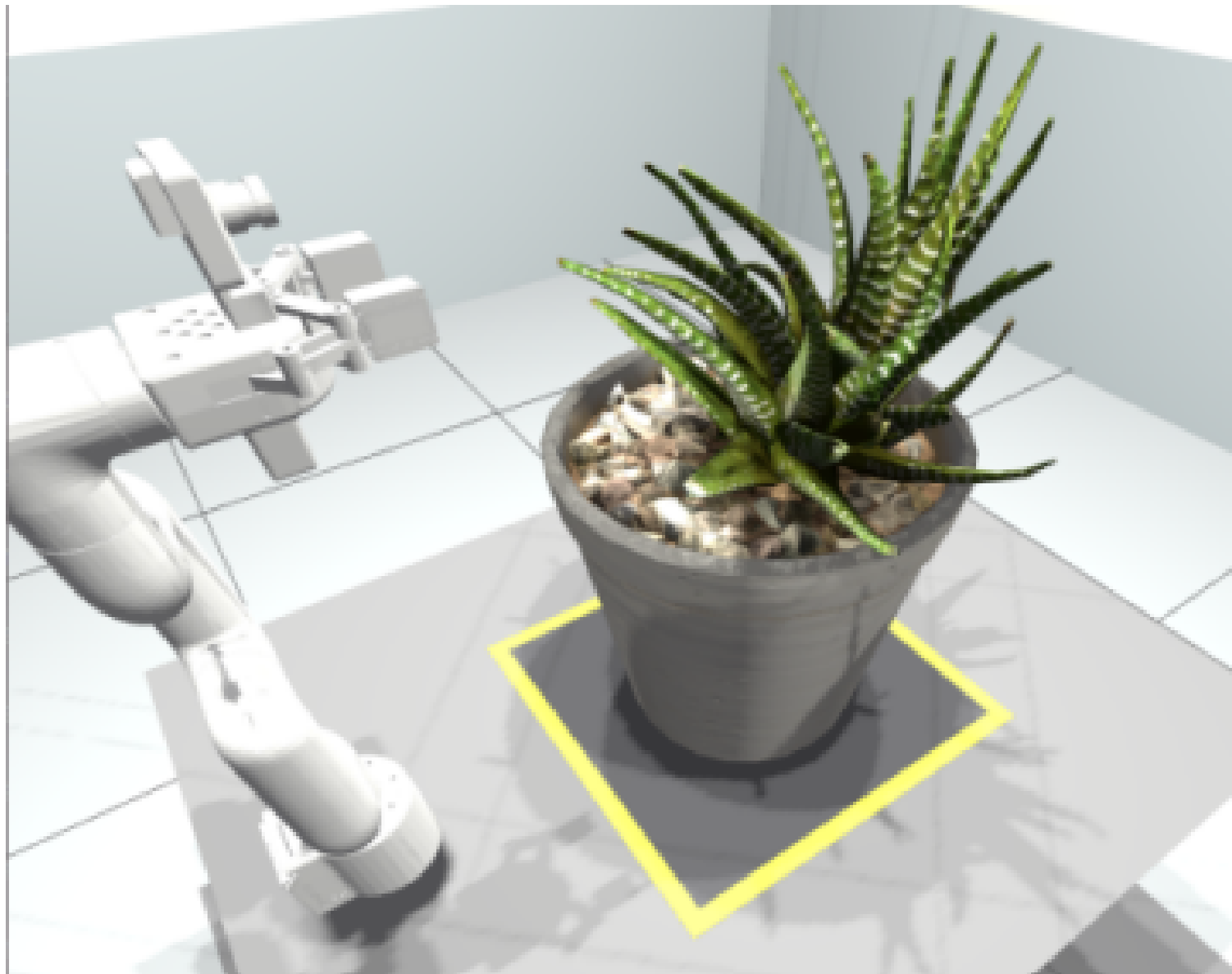


These interconnected steps form my pipeline, all operating on top of ROS 2 Jazzy running on Ubuntu Linux

ROS 2 Control & Simulation Stack



ROS 2 Control & Simulation Stack



3D Reconstruction Algorithms

1

Save Acquisition Data

Acquisition & Pose
Logging

2

Pose Estimation

Pose Recovery (SfM vs.
Learning)

3

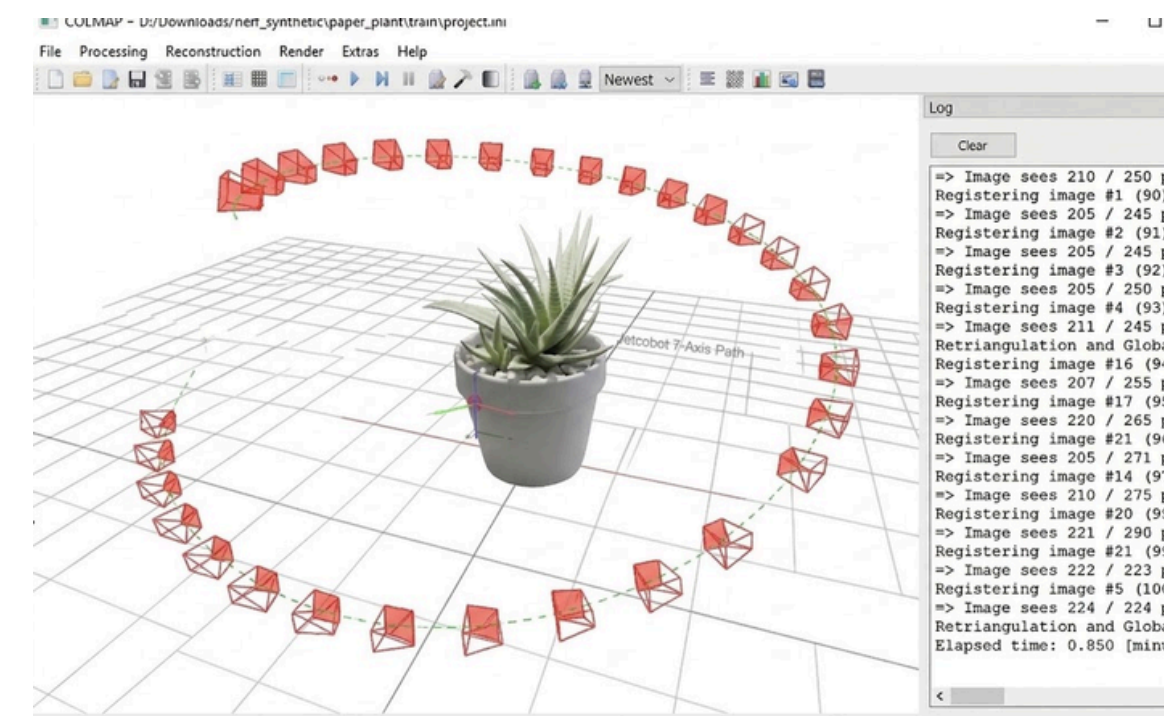
3D Scene
Representation

Neural Rendering (NeRF &
3DGS)

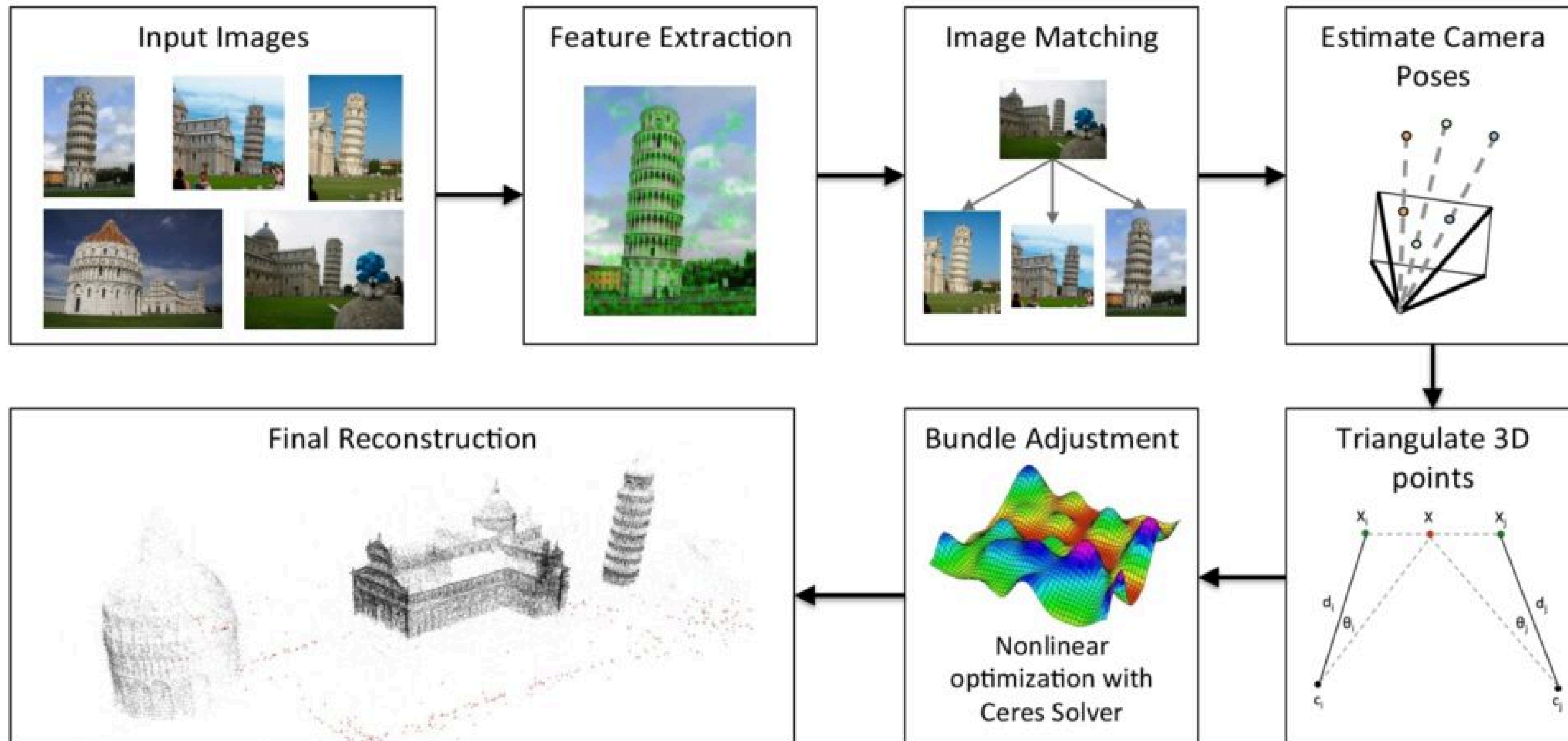
4

Reconstruction
Evaluation

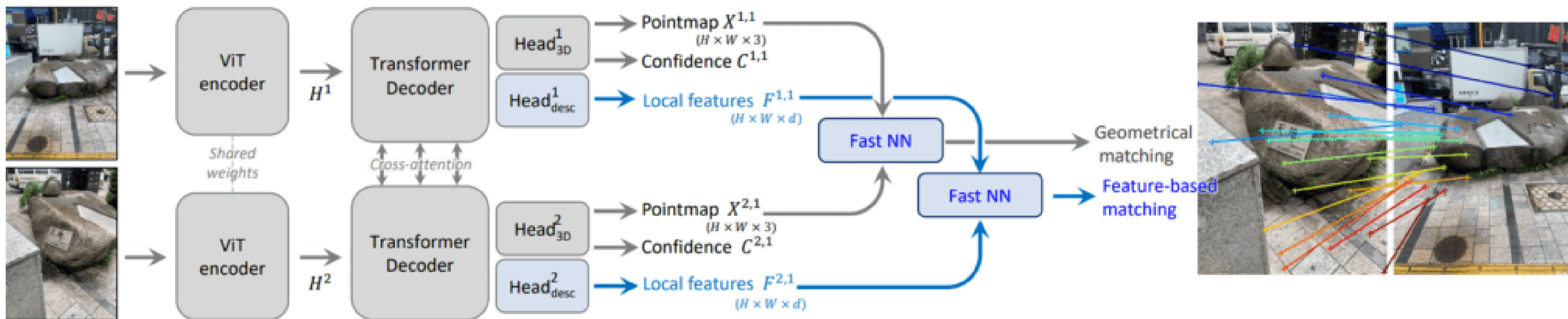
Trajectory & Render
Benchmarking



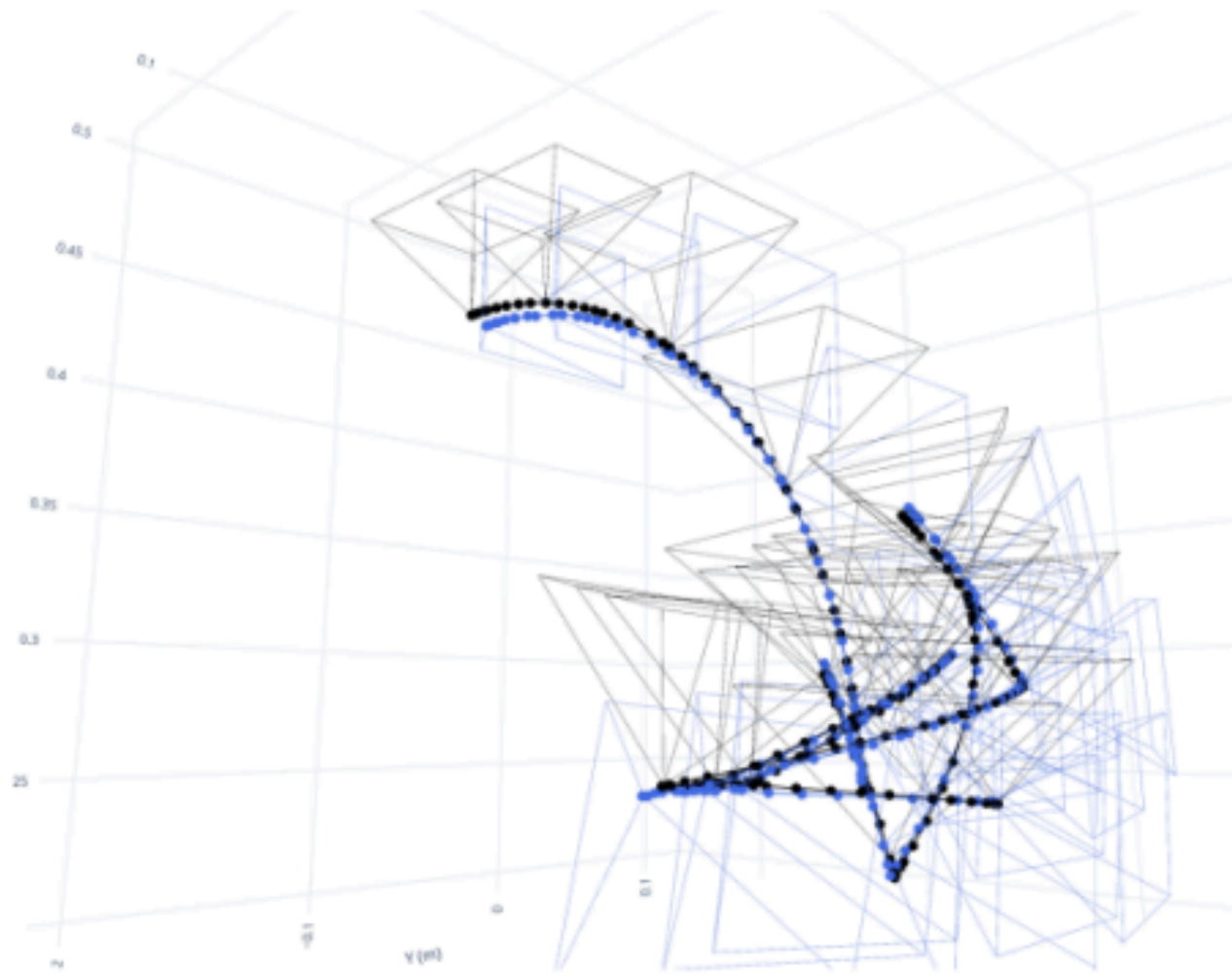
Pose Estimation : COLMAP



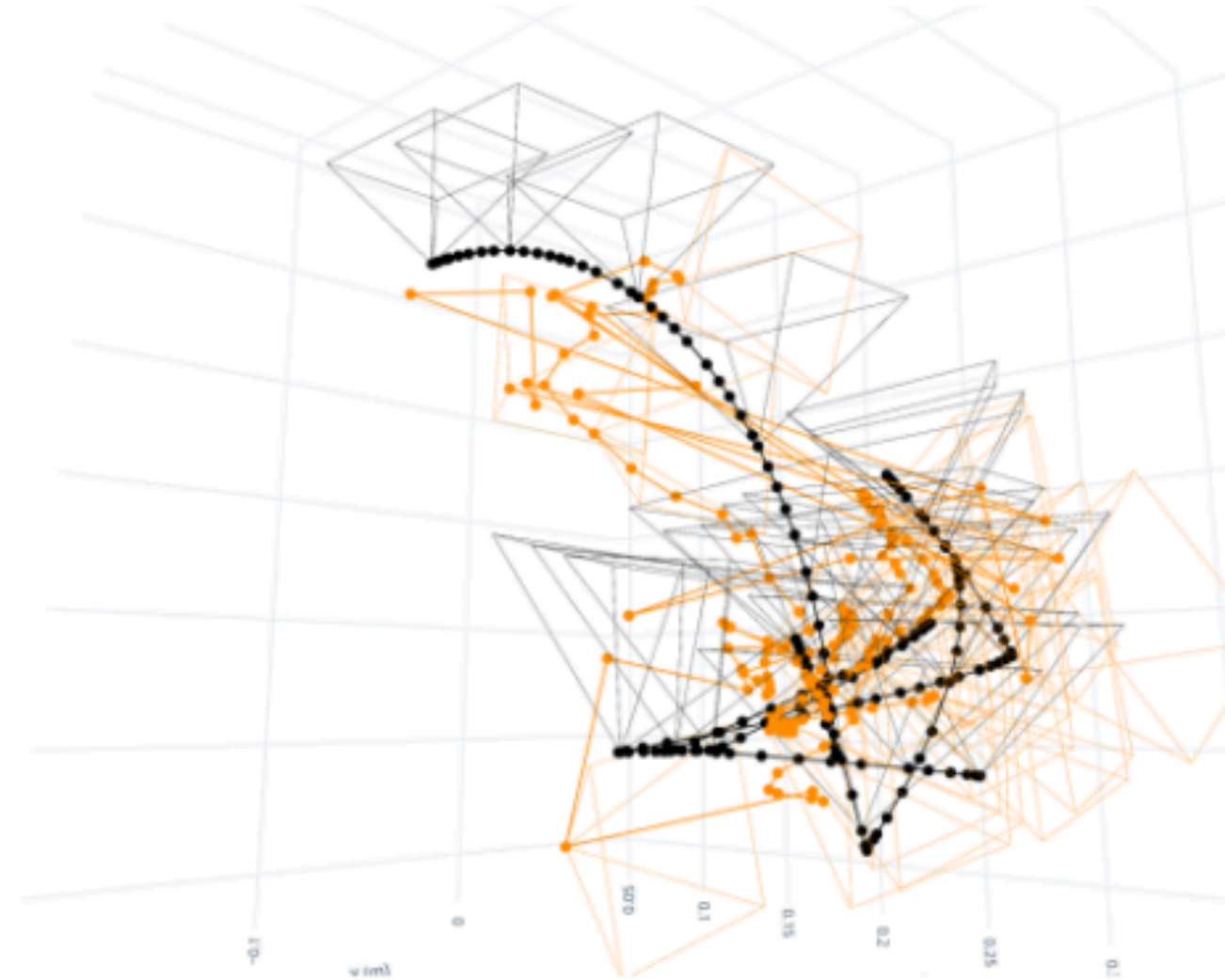
Pose Estimation : MAST3R



Pose Estimation



(a) COLMAP(blue)vs. GT
(black)

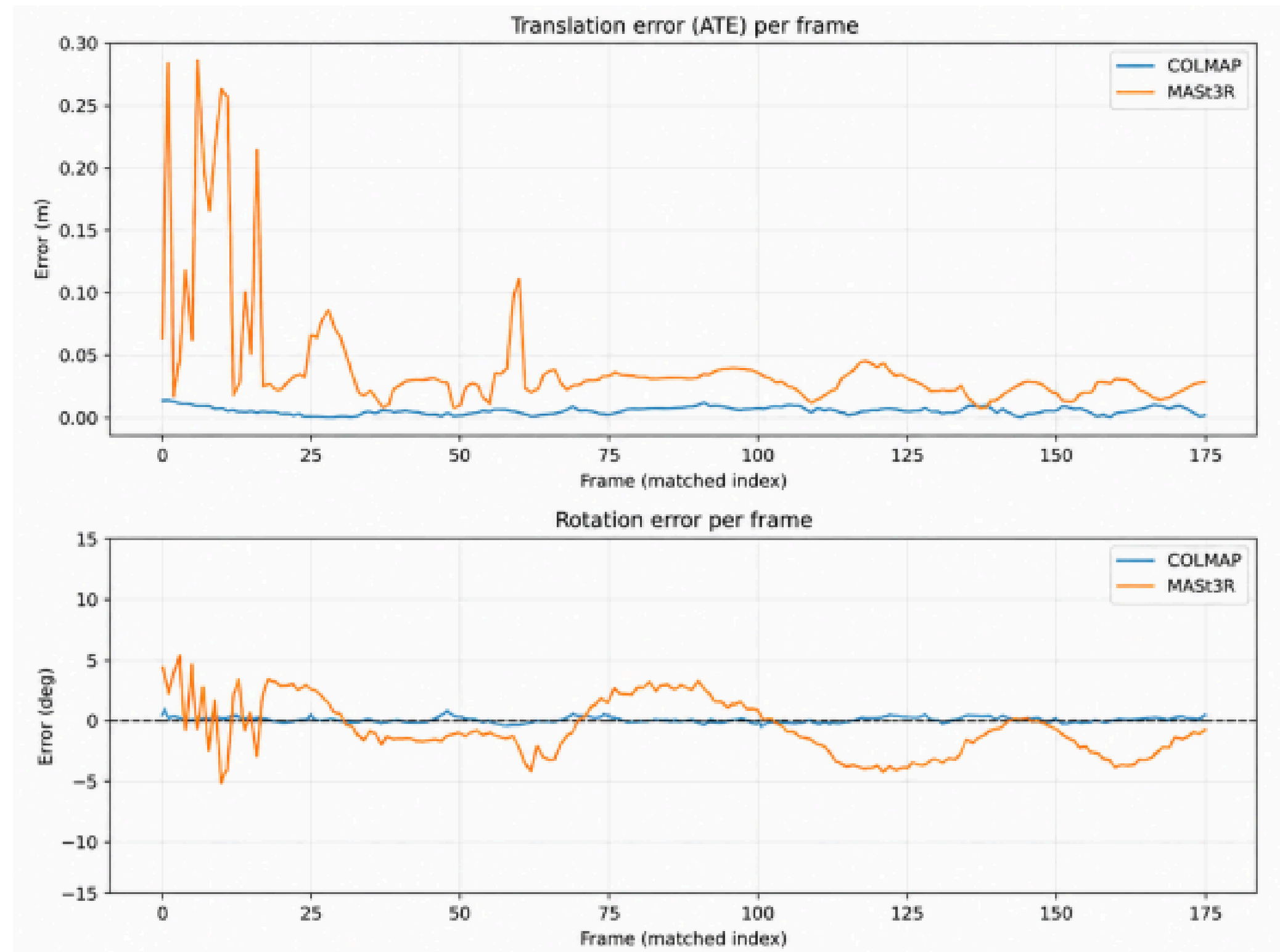


(b) MAST3R(orange) vs.
GT(black)

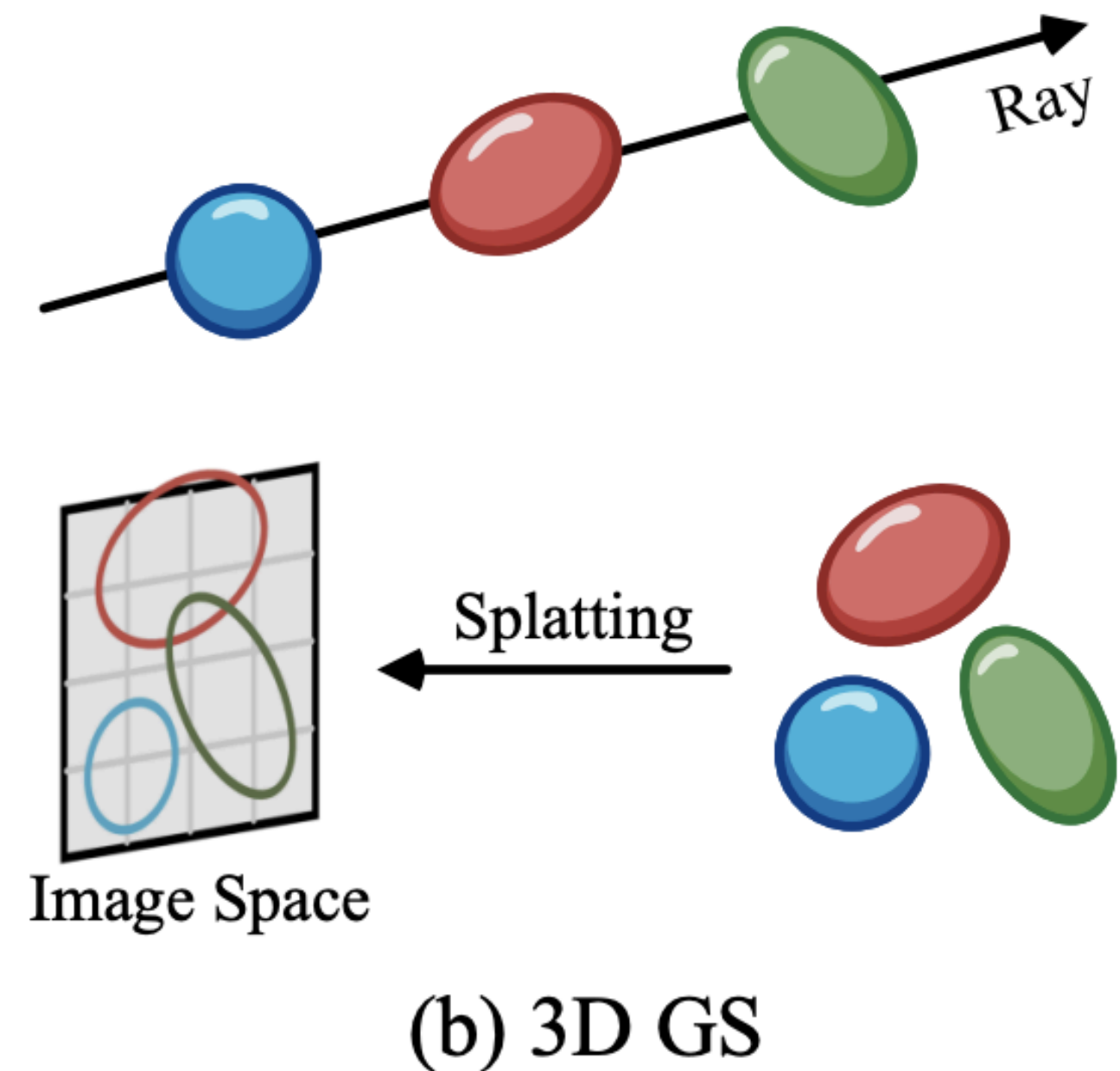
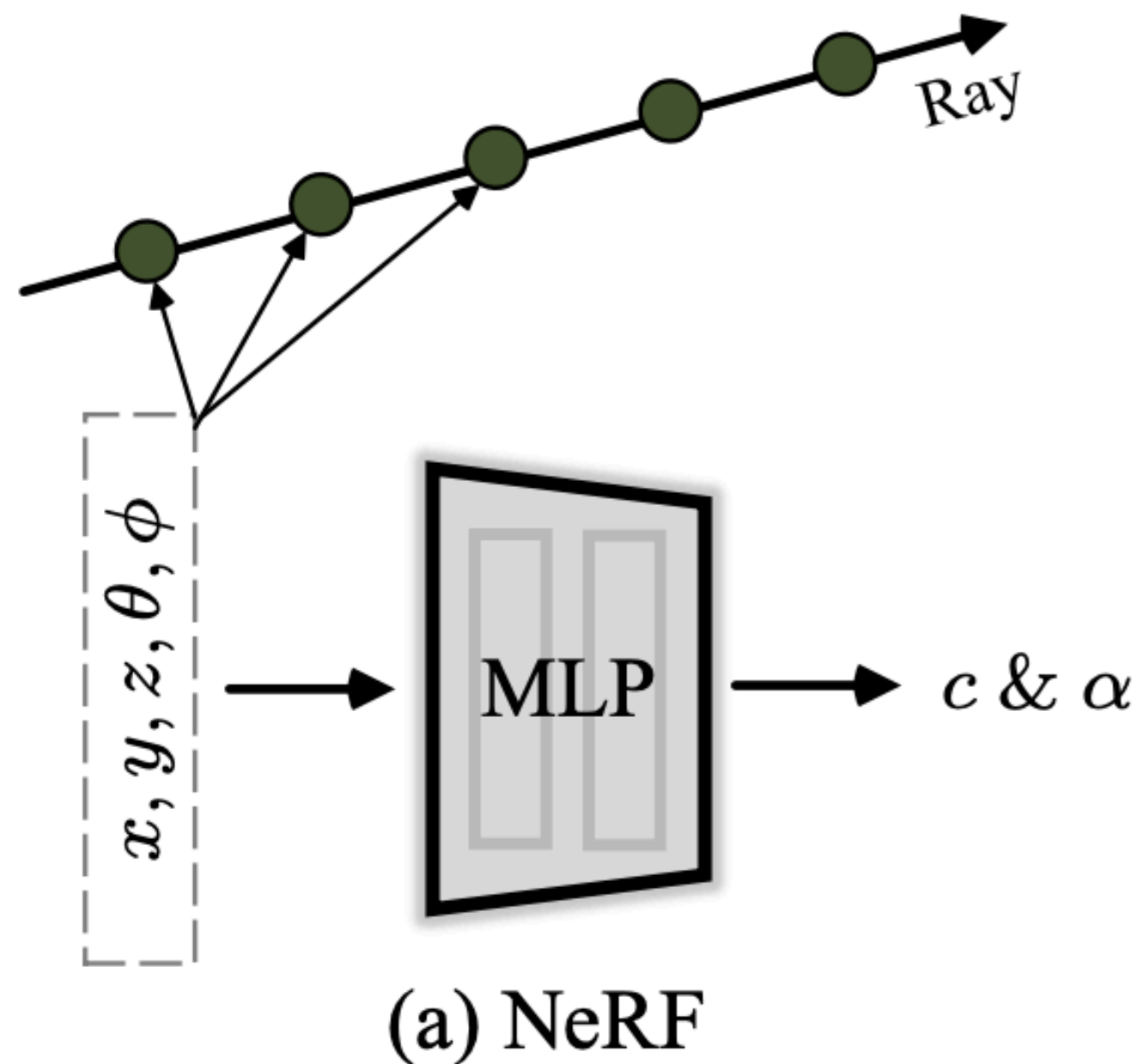
Pose Estimation

$$\text{ATE}_{\text{RMSE}} = \sqrt{\frac{1}{N} \sum_{i=1}^N \|\mathbf{t}_i - \hat{\mathbf{t}}_i\|_2^2}.$$

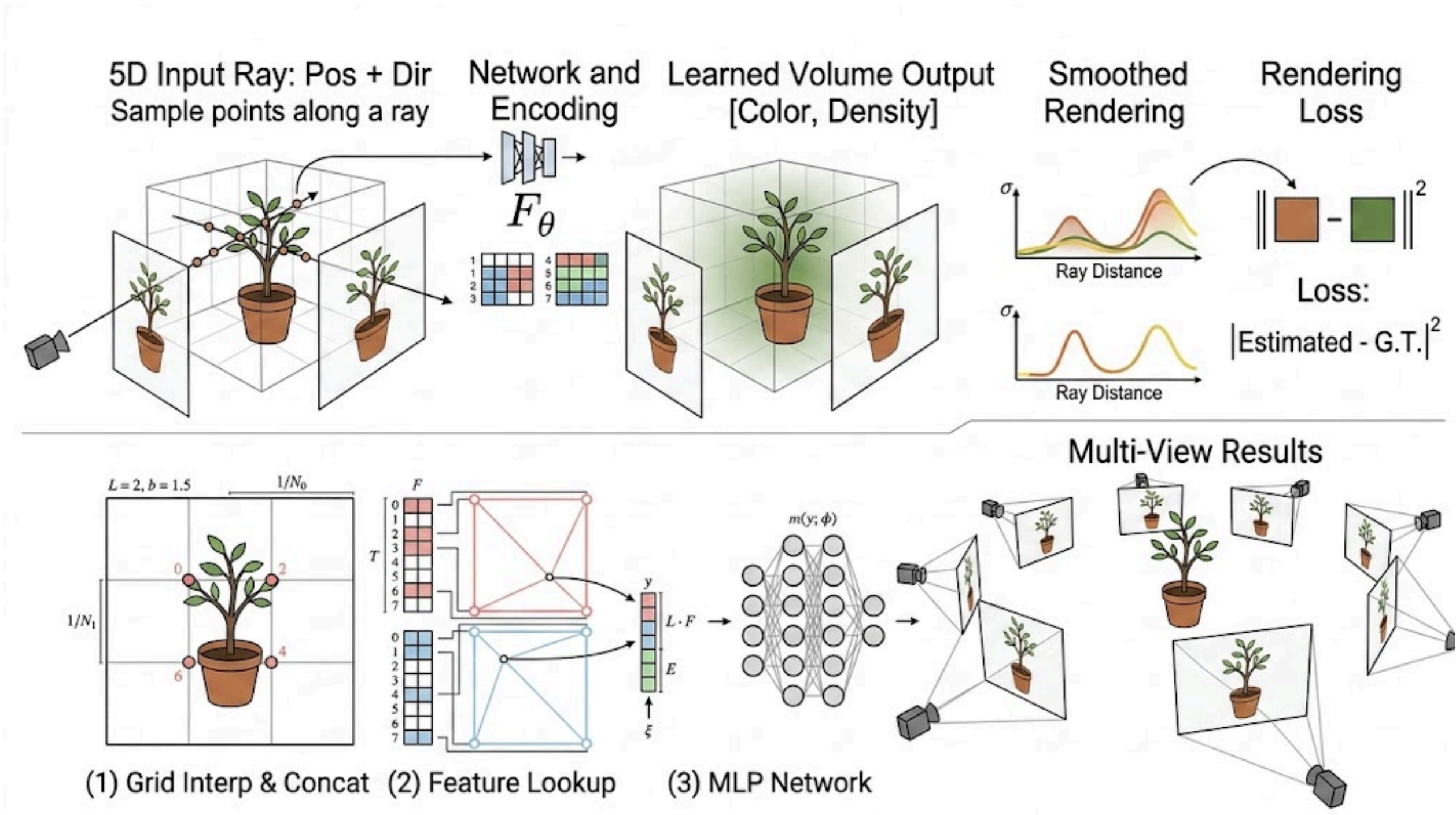
$$e_R^{(i)} = \cos^{-1} \left(\frac{\text{tr} \left(\mathbf{R}_i^T \hat{\mathbf{R}}_i \right) - 1}{2} \right) \frac{180}{\pi}$$



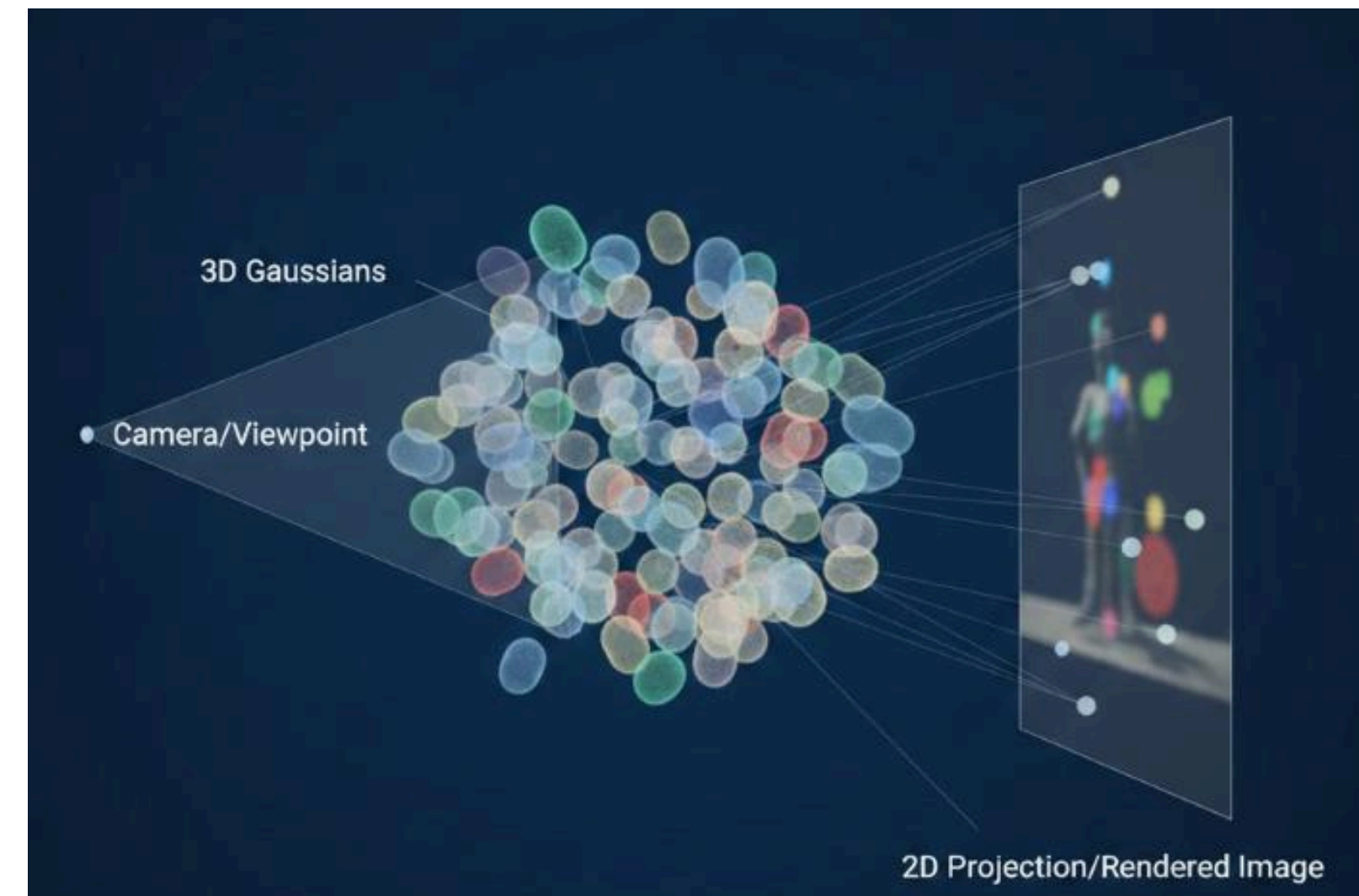
Reconstruction 3D Scene Representation : NERF & 3D-GS



Reconstruction 3D Scene Representation : NERF

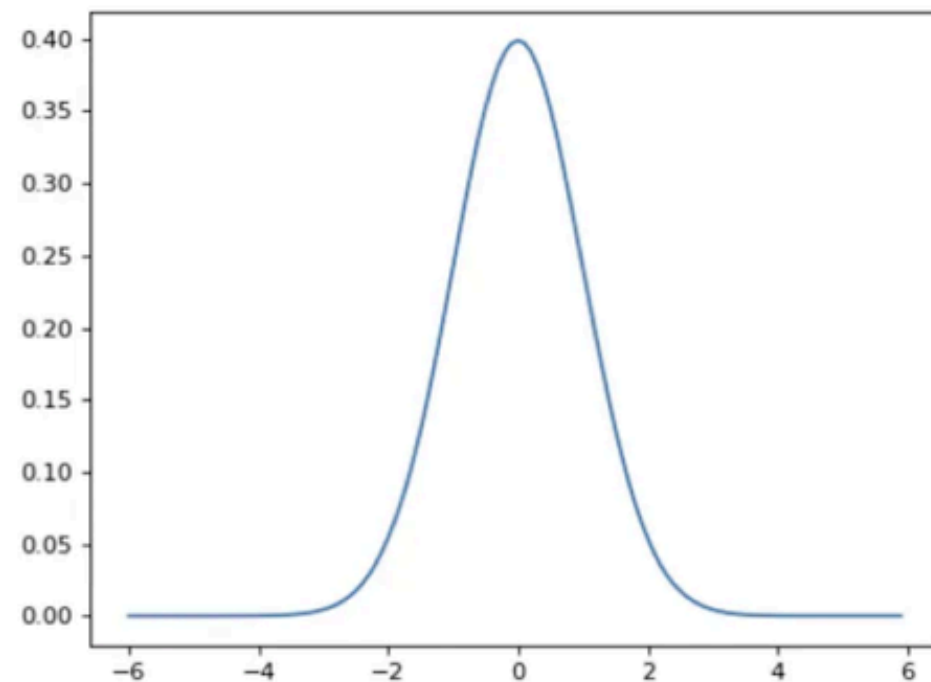


Reconstruction 3D Scene Representation : Gaussian

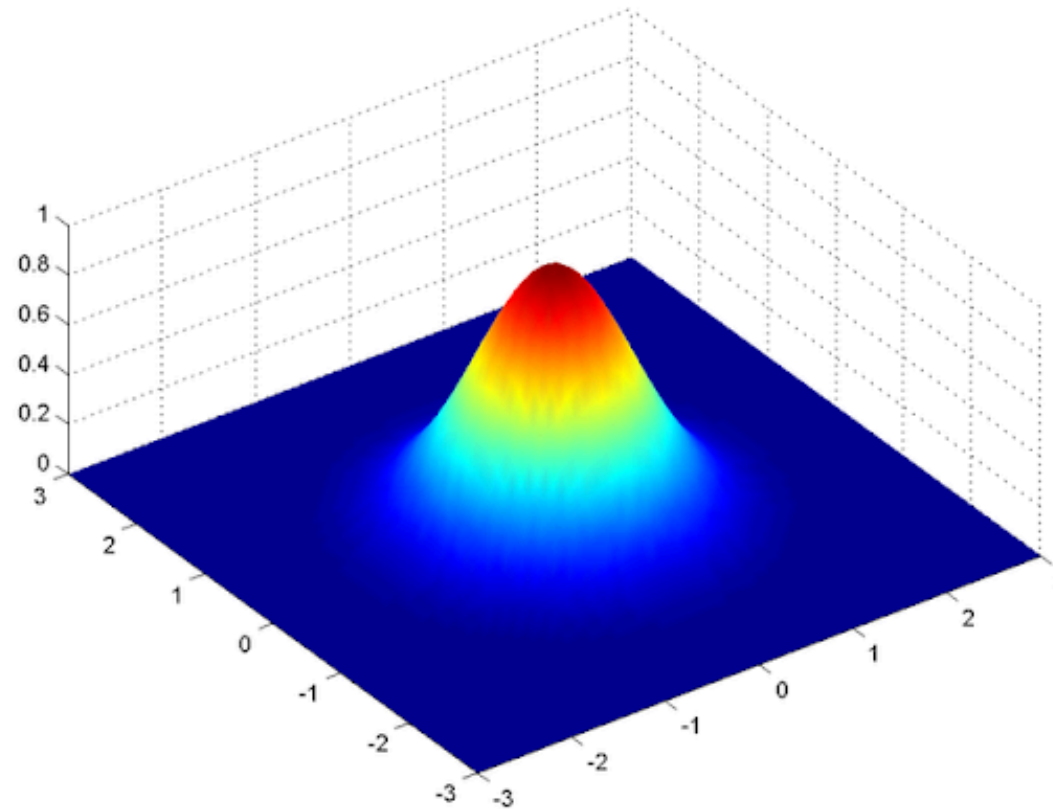


Reconstruction 3D Scene Representation : Gaussian

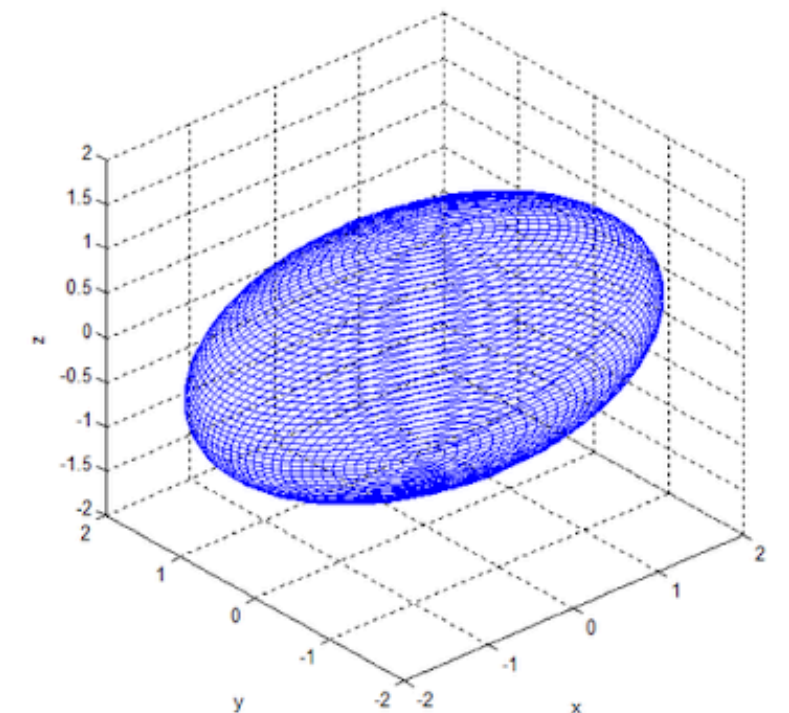
1D Gaussian



2D Gaussian

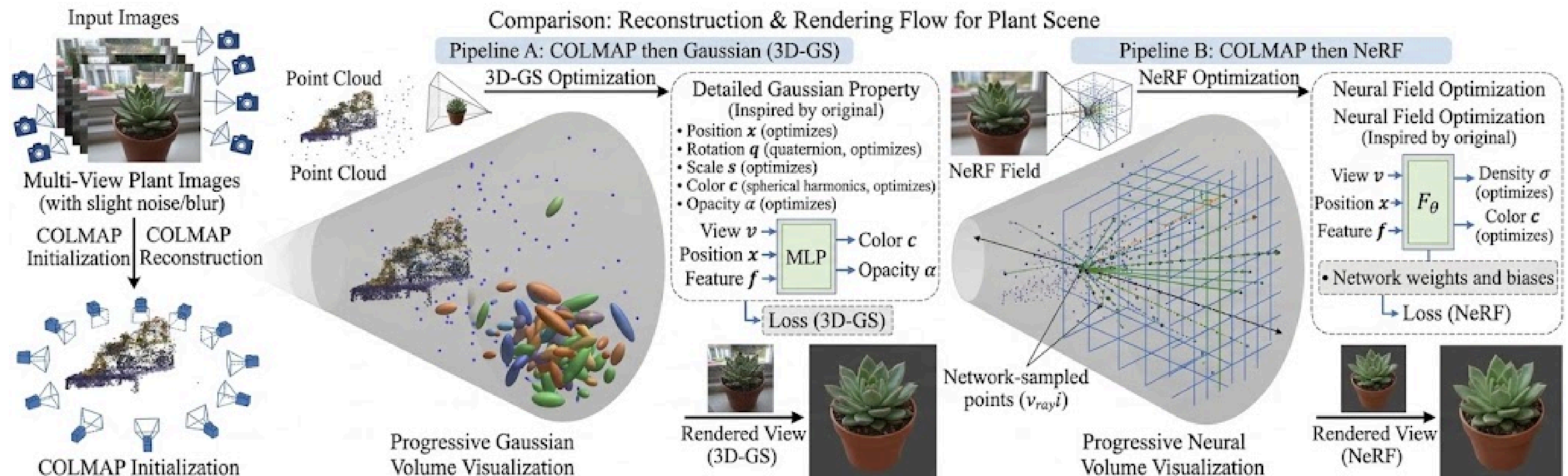


3D Gaussian



Position + covariance + opacity + color

Reconstruction 3D Scene Representation : NERF & 3D-GS



Reconstruction 3D Scene Representation and Evaluation Results

Metric	NeRF (COLMAP)	NeRF (MASt3R)	GS (COLMAP)
PSNR $\uparrow$	19.93	14.17	19.00
SSIM $\uparrow$	0.79	0.55	0.85
LPIPS $\downarrow$	0.16	0.42	0.11

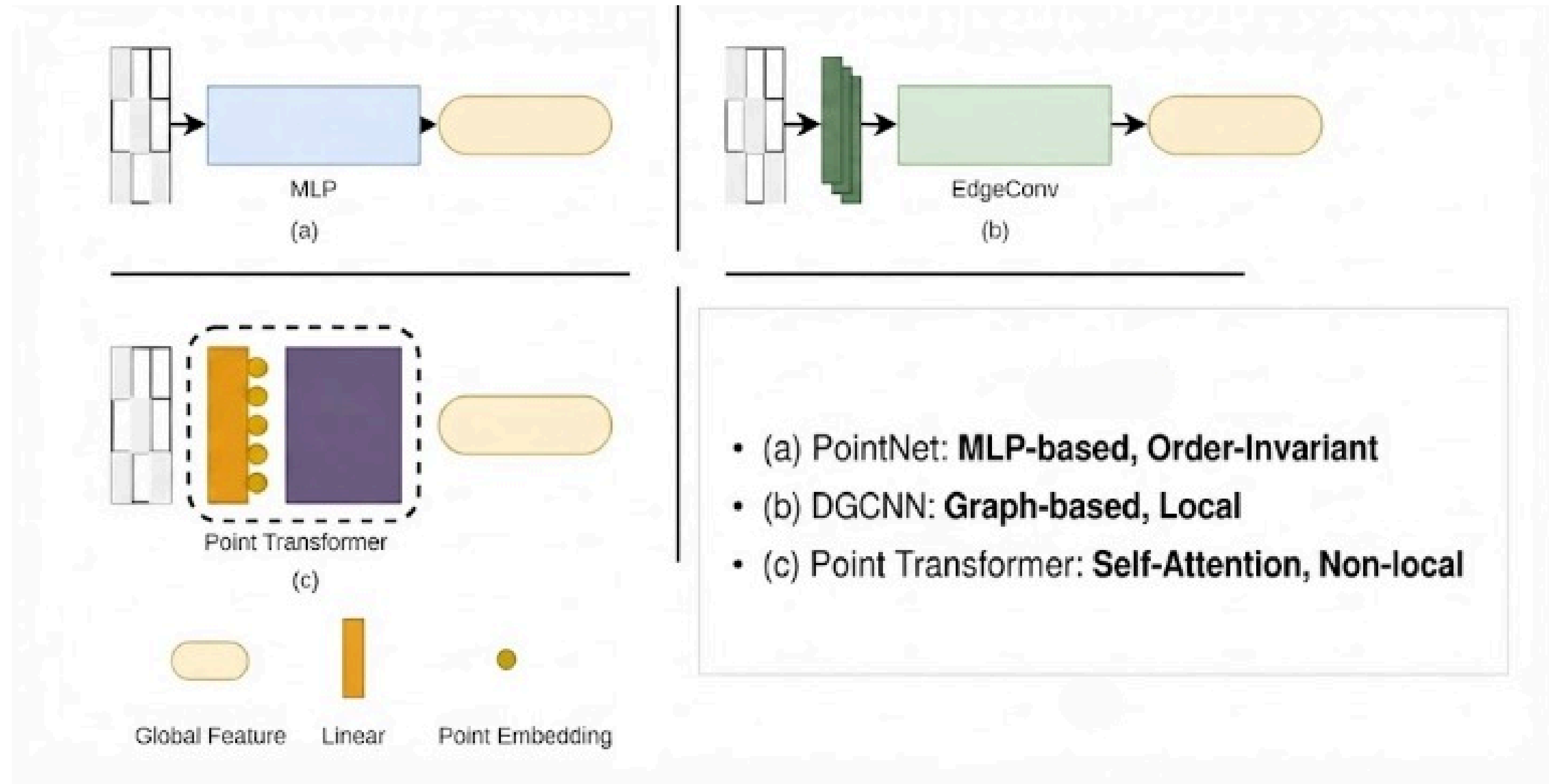


PSNR $\uparrow$ — Measures pixel-level reconstruction quality; higher is better.

SSIM $\uparrow$ — Measures structural similarity between images; higher is better.

LPIPS $\downarrow$ — Measures perceptual difference between images; lower is better.

3D Point Cloud Segmentation algorithms



3D Point Cloud Segmentation results

$$\text{IoU} = \frac{TP}{TP + FP + FN},$$

where TP , FP , and FN denote the numbers of true-positive, false-positive, and false-negative points, respectively.

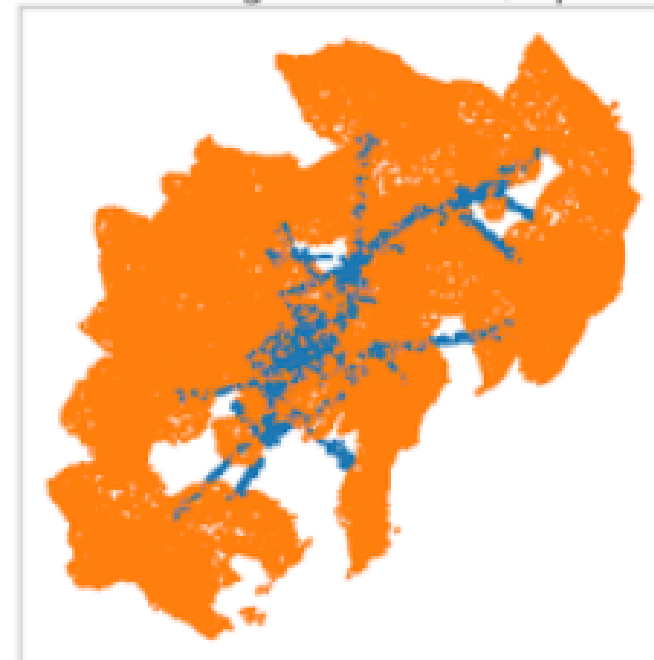
The mean Intersection over Union (mIoU) over C classes is defined as

$$\text{mIoU} = \frac{1}{C} \sum_{c=1}^C \text{IoU}_c,$$

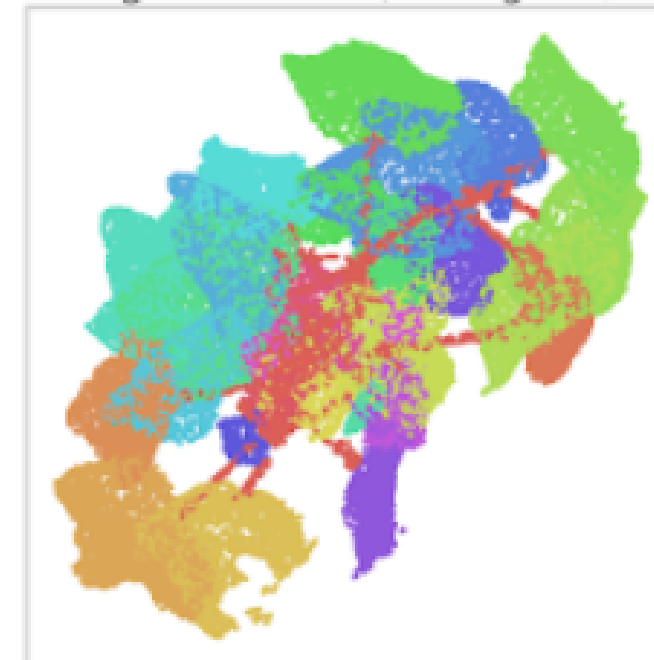
where C denotes the total number of evaluated classes, c is the class index, and IoU_c denotes the IoU for class c .

Model	mIoU	Accuracy
PointNet++	0.9064	0.9886
PTv3	0.9354	0.9829
DGCNN	0.6205	0.7652

Semantic segmentation (top view)



Instance segmentation (33 organs, top view)



3D Point Cloud Segmentation results

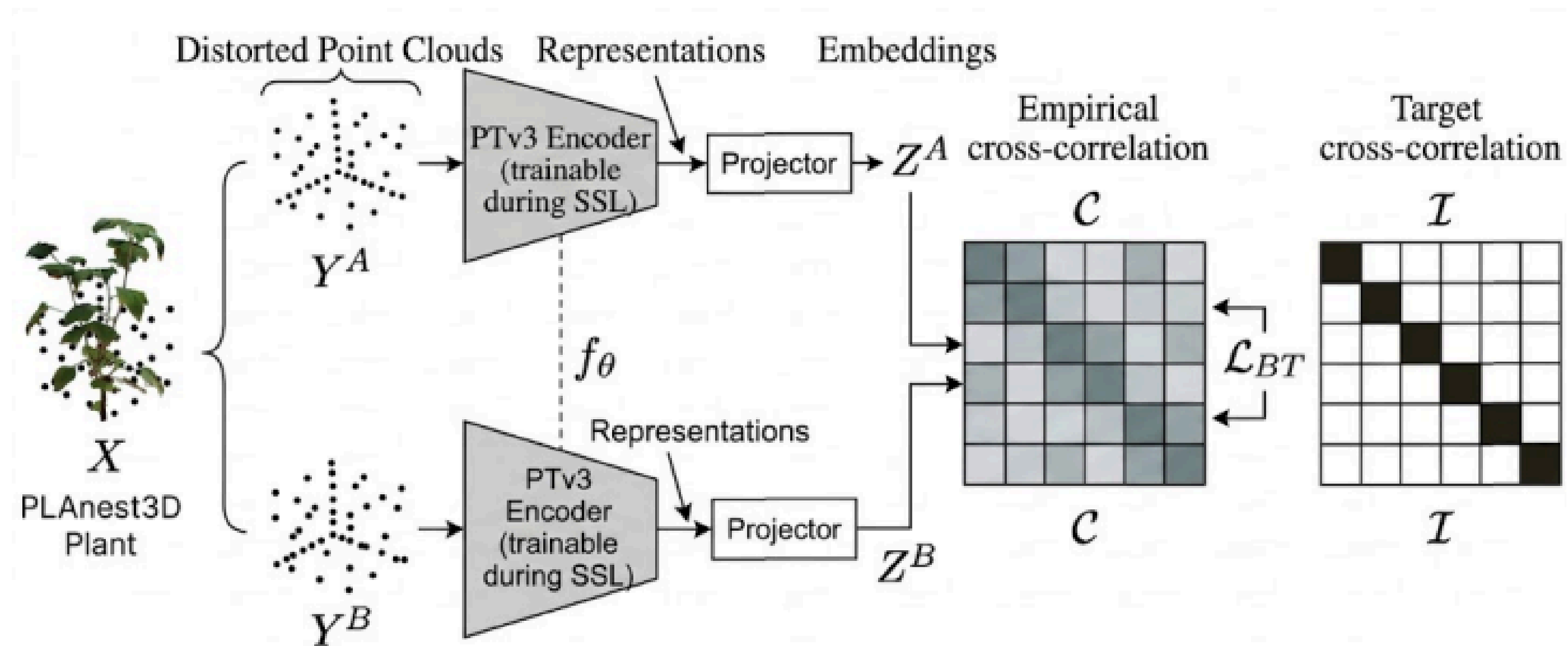
Model	GT mIoU	Reconstructed mIoU	Degradation (%)
PointNet++	0.9064	0.8776	3.18
DGCNN	0.6205	0.5916	4.66
PTv3	0.9354	0.8731	6.65

$$\Delta_{\text{mIoU}} = \text{mIoU}_{\text{GT}} - \text{mIoU}_{\text{reconstructed}},$$

with the relative degradation given by

$$D_{\text{mIoU}} = \frac{\Delta_{\text{mIoU}}}{\text{mIoU}_{\text{GT}}} \times 100.$$

Self-Supervised Adaptation : barlow twins and Utonia



Barlow Twins : Loss function

$$\mathcal{L}_{BT} = \underbrace{\sum_i (1 - C_{ii})^2}_{\text{Invariance}} + \lambda \underbrace{\sum_{i \neq j} C_{ij}^2}_{\text{Redundancy reduction}}$$

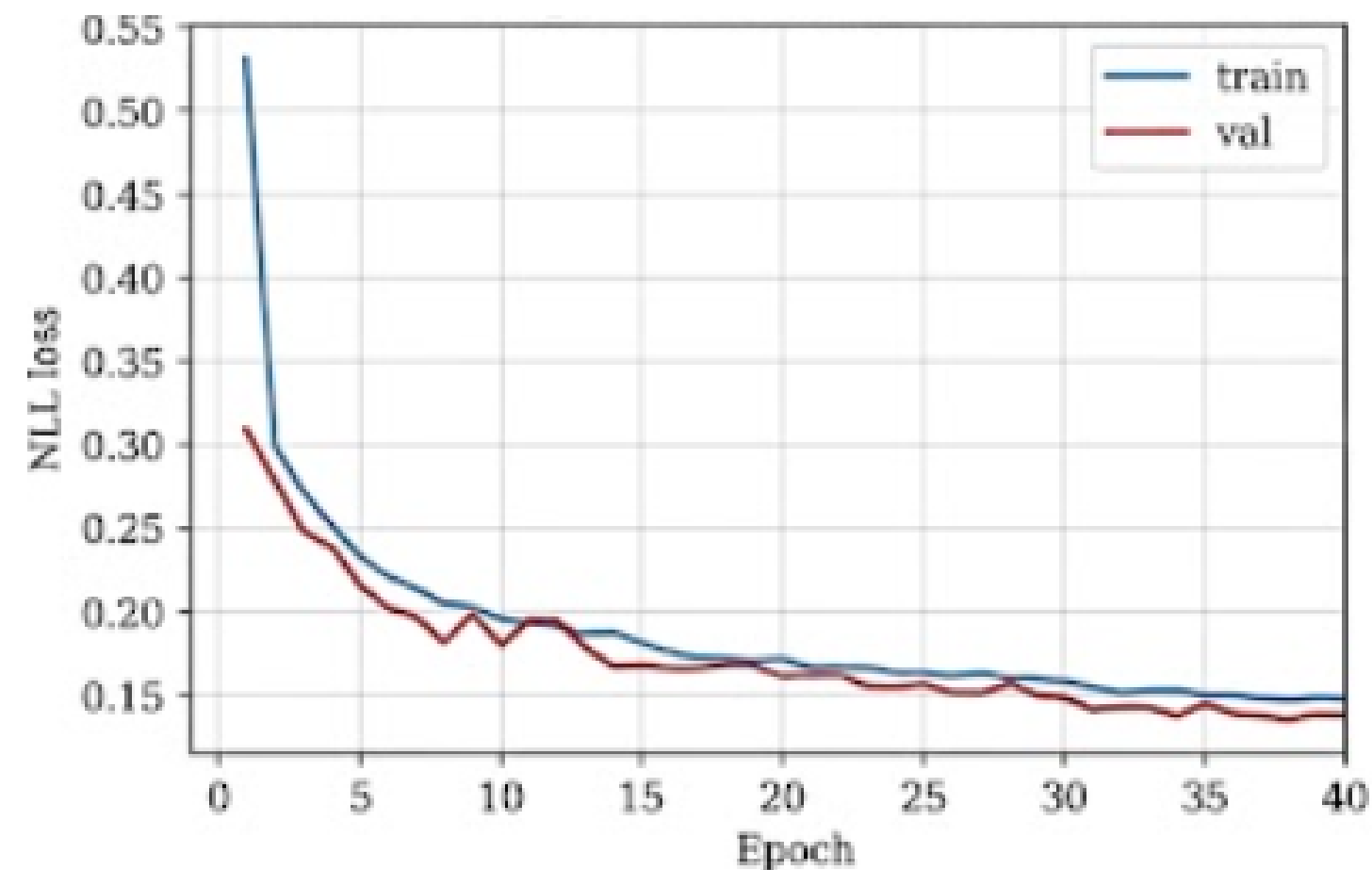
where $C = \frac{1}{B} \tilde{Z}_{GT}^T \tilde{Z}_{REC}$ is the cross-correlation matrix, B =batch size, and λ =balance factor.

Goal: $C \rightarrow I \rightarrow$ invariant and non-redundant representations.

$$C_{ij} \triangleq \frac{\sum_b z_{b,i}^A z_{b,j}^B}{\sqrt{\sum_b (z_{b,i}^A)^2} \sqrt{\sum_b (z_{b,j}^B)^2}}$$

Barlow Twins : Results

Method	GT mIoU (%)	Δ (pp)	REC mIoU (%)	Δ (pp)
PTv3 supervised	93.54	–	87.31	–
PTv3 + Utonia	88.76	–4.78	87.81	+0.50
PTv3 + Barlow Twins	89.50	–4.04	89.03	+1.72



Conclusion

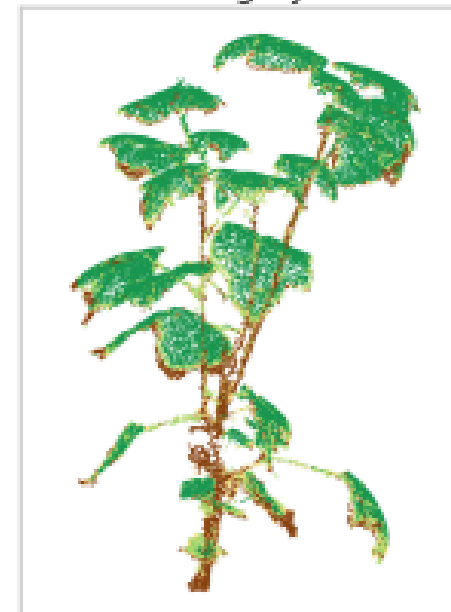
This work presents a robotic RGB scanning pipeline uses self-supervised Barlow Twins adaptation to overcome reconstruction noise and enhance segmentation —enabling precise downstream organ attribution, skeletonization, and health analysis .



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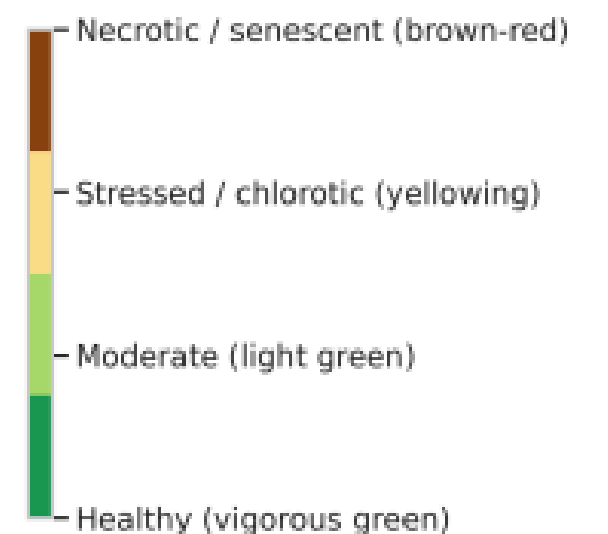
Health category — Front



Health category — Side



Health category — Top



Thank You! — Questions & Discussion



